

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} g_v^+ &= v\psi, (\text{resp. } g_v^- = v\psi), \\ r_v &= v\psi_u, \\ \psi_v \cdot \psi &= \phi_v \cdot \phi, \\ \psi_v \cdot g_u^+ &= \phi_v \cdot f_u (\text{resp. } \psi_v \cdot g_u^- = \phi_v \cdot f_u), \\ \psi_v \cdot r_u &= (\phi \cdot f_{uu})_v - \frac{v}{2}(\phi \cdot \phi)_{uu} + v(\psi_u \cdot \psi_u), \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\mathbb{D}_\lambda^2 i. [(\nabla_\lambda i)^{-T} \nabla_\lambda g] = -k e^{\frac{k}{2} \log^2 \frac{\lambda_1}{\lambda_2}} \left(\log \frac{\lambda_1}{\lambda_2} \right) \frac{1}{(\lambda_1 - \lambda_2)} \left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2} \right) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &Q_1(f, g) - Q_1(h, h_-) + Q_3(f, g) - Q_3(h, h_-) \\ &= Q_1(f - h, g - h_-) + Q_1(f, h_-) + Q_1(g, h) - 2Q_1(h, h_-) \\ &+ Q_3(f - h, g - h_-) + Q_3(f, h_-) + Q_3(g, h) - 2Q_3(h, h_-) \\ &\stackrel{(??.)}{=} Q_1(f - h, g - h_-) + Q_3(f - h, g - h_-). \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \frac{d}{d\lambda} S_{\omega, \mathbf{c}}(\Phi^\lambda)|_{\lambda=1} &= \frac{d}{d\lambda} \left(L(\Phi^\lambda) + N(\Phi^\lambda) + \omega Q(\Phi^\lambda) + \mathbf{c} \cdot \mathbf{P}(\Phi^\lambda) \right) \Big|_{\lambda=1} \\ &= 2L(\Phi) + \left(\frac{d}{2} + 1 \right) N(\Phi) + \mathbf{c} \cdot \mathbf{P}(\Phi) \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \Phi^{\iota_1 \iota_2 \iota_3}(\xi, x, y) &:= -\Lambda(\xi) + \iota_1 \Lambda(\xi + x) + \iota_2 \Lambda(\xi + y) + \iota_3 \Lambda(\xi + x + y), \\ c_{\xi; k_1, k_2, k_3}^{*, \iota_1 \iota_2 \iota_3}(x, y) &:= c^{\iota_1 \iota_2 \iota_3}(\xi, -x, -\xi - x - y) \cdot \varphi'_{k_1}(\xi + x) \varphi'_{k_2}(\xi + y) \varphi'_{k_3}(\xi + x + y). \end{aligned}$$